Capacity-Radius Proximity for Smooth-Domain Reed–Solomon Codes

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Abstract

We study list decoding and same-set mutual correlated agreement for Reed–Solomon codes over smooth evaluation domains. A two-coordinate plane is exactly an interleaved pair word, equivalently the base–direction pair of a scalar affine line.

At count radius $n-k$, an agreement-set injection and a matching received-word construction prove

$$L_m\left(\mathcal{C}, \frac{n-k}{n}\right) = \binom{n}{k}$$

for $1 \le k \le n$ and every positive interleaving width. A next-shell construction gives the three-regime fractional-list phase split. For pair mutual correlated agreement, compressed interpolation certificates give

$$|\,\mathrm{Bad}_{\mathrm{MCA}}\,| \le \binom{n}{k+1}$$

at every radius. A smooth rate-1/4 family over $\mathbb{F}_{3^n}$ meets both 2^{-128} fractional budgets: its list target holds at capacity and its pair-MCA target holds at every normalized radius. For the separate prime-field instance $(q, n, k) = (2^{128} + 192513, 4096, 1024)$, the exact bad-count spectrum has $B_{\mathcal{C}}(0) = 1$ and $B_{\mathcal{C}}(1) \ge 2$. Its greatest safe MCA count radius is zero, its first unsafe count is one, and its real safe set is $[0, 1/4096)$. The intermediate field-size list optimum remains open.

For the shared systematic cellular cochain construction, we prove a quantitative comparison for its existing committed-cochain projection and intrinsic modular coboundary validity. Its independently defined native distances satisfy $d_C \le d_P = D_2 \le 2d_C$, and a well-formed raw payload in the canonical $q = 12289$ instance attains 3/4 at all three levels. This application uses a separate field from the two fractional-budget instances above. Euclidean and full-HCP source comparisons remain separate.

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1 Introduction

The Grand List Decoding and Grand Mutual Correlated Agreement challenges ask for quantitative proximity guarantees for constant-rate Reed–Solomon codes over smooth evaluation domains [ABF26]. We work with the two-fold interleaving $\mathcal{C}^{\equiv 2} \subseteq (\mathbb{F}^2)^n$. Every received pair $p = (p_0, p_1)$ determines the scalar affine line

$$p \in (\mathbb{F}^2)^n \quad \longmapsto \quad y_\gamma = p_0 + \gamma p_1 \in \mathbb{F}^n.$$

List decoding controls the ambiguity of the full pair, while mutual correlated agreement bounds the fraction of challenges for which scalar agreement fails to extend to pair agreement on the same coordinate set.

At the capacity-grid radius $(n-k)/n$, we prove the exact universal list size

$$L_m\left(\mathcal{C}, \frac{n-k}{n}\right) = \binom{n}{k}$$

for every positive interleaving width. A matching next-shell construction gives the resulting three-regime field-size phase split. For pair MCA, an injective compressed-certificate map proves the radius-independent bound

$$|\,\mathrm{Bad}_{\mathrm{MCA}}\,| \le \binom{n}{k+1}.$$

An explicit smooth rate-1/4 family over $\mathbb{F}_{3^n}$ meets both 2^{-128} fractional targets. Its actual field/subgroup realization, endpoint constructions, MCA certificate bound, complete phase split, real safe intervals and generic distance/calibration endpoints are machine-checked in Lean. The endpoint arguments are elementary interpolation; the exponential alphabet and allowable lists do not establish an efficient decoder or a polynomial-list improvement.

Recent limitations clarify this distinction. Alrabiah, Goyal and Guruswami [AGG26, Theorems 1.1, 1.2 and 1.4] construct super-polynomial lists just beyond Johnson for certain full-length low-characteristic codes, and growing polynomial lists below capacity for specified full-length prime-field codes. Their full length is the field cardinality. Our family instead uses n

subgroup points in $\mathbb{F}_{3^n}$, and its endpoint list $\binom{n}{k}$ is allowed to grow exponentially in n while meeting a fraction-of-field budget. Neither that budget nor the separate pair-MCA event asserts polynomial-list decoding up to capacity.

To resolve the first nontrivial MCA transition rather than only give a uniform sufficient bound, we define the exact spectrum

$$B_{\mathcal{C}}(r) = \max_{p} |\operatorname{Bad}_{\mathrm{MCA}}(\mathcal{C}, r/n, p)|.$$

For the certified smooth prime-field instance $q = 2^{128} + 192513 \equiv 1 \pmod{4096}$, $n = 4096$, and $k = 1024$, we prove $B_{\mathcal{C}}(0) = 1$ and $B_{\mathcal{C}}(1) \geq 2$. It follows that zero is the greatest safe count radius and one the first unsafe count radius at the 2^{-128} MCA error target. Over real radii the safe interval is $[0, 1/4096)$, so its supremum is $1/4096$ and is not itself safe.

The coding theorems quantify over the complete ambient received-word and affine-line spaces. Their 2^{-128} instantiation uses the displayed exponential-size field and is not a claim about every supplied smooth code. An application with an independently defined source metric additionally requires a comparison with the extracted pair-code distance.

The source application uses the cochain construction and observation profile of [vdV26b]. Its HCP and cellular Hodge primitives are those of [vdV26a]; the projection, reconstruction and source-distance comparisons are proved here. The coding theorems do not assume an HCP presentation or a commitment protocol. A protocol using them must enforce the fixed-pair, later-challenge order described in Section 6. Section 7 uses Paper 2's explicit systematic cellular construction and proves its source-distance comparison. It retains the original committed-cochain payload in a complete coordinate layout and defines source validity intrinsically by an integral opening and carry satisfying $\delta_0 a = z + qm$. Scalar cochain edits count $2n$ coordinates, while plane and RS edits count n whole pairs. The proved global comparison is

$$d_C(z) \leq d_P(z) = D_2(\bar{z}) \leq 2d_C(z).$$

The factor two is necessary in the near direction; the far implication has unit scale. An actual well-formed raw payload at $(q, n, k) = (12289, 4096, 1024)$ has all three distances equal to $3/4$. This is a modular source theorem. The Euclidean and full-HCP residuals have different comparison obligations, and positive-far raw payloads are distinct from already accepted exact storage anchors. The selected application field does not inherit the large-field family's 2^{-128} guarantee.

2 Problem Setup

2.1 Reed–Solomon parameters

Let $\mathbb{F}$ be a finite field and let $\mathcal{L} = (a_i)_{i \in I}$ be an injectively indexed evaluation domain, where $|I| = n$. For $1 \le k \le n$, define

$$\mathcal{C} = \mathrm{RS}[\mathbb{F}, \mathcal{L}, k] = \{(f(a_i))_{i \in I} : f \in \mathbb{F}[X],\ \deg f < k\}\,.$$

The rate is $\rho = k/n$. The smooth-domain regime takes ρ from a fixed constant set, such as

$$\left\{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}\right\},$$

and requires $\mathcal{L}$ to have an appropriate smooth structure.

2.2 Relative Hamming distance

For a finite alphabet A and $u, v \in A^I$, let

$$d_H(u, v) = |\{i \in I : u_i \neq v_i\}|, \qquad \Delta(u, v) = \frac{d_H(u, v)}{n}.$$

For a nonempty code $\mathcal{D} \subseteq A^I$, define

$$\Delta(w, \mathcal{D}) = \min_{c \in \mathcal{D}} \Delta(w, c).$$

Unless stated otherwise, radii are relative radii in $[0, 1]$. The uniform MCA bound also allows arbitrary real values of its radius parameter. Natural-number radii describe the finite Hamming grid and its rounding; distances in the main statements are normalized by n.

2.3 Interleaving

For a positive integer m, define

$$\mathcal{C}^{\equiv m} = \left\{p \in (\mathbb{F}^m)^I : \text{for every } j \in \{1, \dots, m\},\ (p_{i,j})_{i \in I} \in \mathcal{C}\right\}.$$

Hamming distance on $(\mathbb{F}^m)^I$ counts a coordinate i as erroneous when the complete vector symbol differs. It does not sum componentwise scalar distances.

2.4 List-decoding balls

For $w \in (\mathbb{F}^m)^I$ and $\delta \in [0,1]$, define

$$\Lambda(\mathcal{C}^{\equiv m}, w, \delta) = \{c \in \mathcal{C}^{\equiv m} : \Delta(w,c) \le \delta\}.$$

The worst-case list size is

$$L_m(\mathcal{C}, \delta) = \max_{w \in (\mathbb{F}^m)^I} |\Lambda(\mathcal{C}^{\equiv m}, w, \delta)|.$$

The maximum over received words is essential. A bound on a structured subset of received words is an application result, not the universal list-decoding statement.

3 Planes, Pair Words, and Affine Lines

3.1 The full ambient pair space

Define a field-valued two-coordinate plane to be a function

$$p : I \longrightarrow (\{0,1\} \longrightarrow \mathbb{F}).$$

By reassociating function spaces, the plane carrier is exactly

$$I \to (\{0,1\} \to \mathbb{F}) \cong (\mathbb{F}^2)^I.$$

Thus the word "plane" introduces no restriction on the received-word space.

Proposition 3.1 (Plane–pair equivalence)**.** *The map sending a plane p to the vector word*

$$i \longmapsto (p(i,0), p(i,1))$$

is a bijection from field-valued planes to $(\mathbb{F}^2)^I$. It preserves and reflects Hamming distance.

Proof. The inverse sends a pair word w to the function

$$(i,j) \longmapsto w_i(j).$$

Both composites are pointwise identities. At coordinate i, the two planes are equal exactly when their complete pair symbols are equal, so the Hamming disagreement sets coincide. □

3.2 Pair words as affine lines

For a plane p, define its base and direction by

$$p_0(i) = p(i, 0), \qquad p_1(i) = p(i, 1).$$

The associated affine line of scalar received words is

$$\ell_p(\gamma) = p_0 + \gamma p_1, \qquad \gamma \in \mathbb{F}.$$

Proposition 3.2 (Plane–line equivalence)**.** *The map $p \mapsto (p_0, p_1)$ is a bijection between field-valued planes and affine-line presentations consisting of an arbitrary base word and arbitrary direction word in $\mathbb{F}^I$.*

Proof. Given an arbitrary pair (f_0, f_1), define

$$p(i, 0) = f_0(i), \qquad p(i, 1) = f_1(i).$$

This construction is inverse to taking base and direction. □

3.3 Challenge evaluation is lossy

For fixed γ, the map

$$(p_0, p_1) \longmapsto p_0 + \gamma p_1$$

is generally many-to-one. Consequently, distance of the pair from the code cannot be inferred deterministically from one fixed challenge without an additional good-challenge hypothesis. MCA supplies a probabilistic statement over a challenge sampled after the pair has been fixed.

4 The Capacity-Radius Regime

4.1 The finite Hamming grid

Let $1 \le k \le n$. The capacity endpoint is already an attainable normalized Hamming radius:

$$r_{\text{cap}} = n - k, \qquad \delta_{\text{cap}} = \frac{n - k}{n} = 1 - \rho.$$

There is therefore no rounding loss at the endpoint. For example, the rate-1/4, length-4096 member has

$$k = 1024, \qquad r_{\text{cap}} = 3072, \qquad \delta_{\text{cap}} = \frac{3}{4}.$$

The endpoint theorem below includes the strictly smaller radius 3071/4096.

There is, however, a distinct granularity issue in the phrase "largest relative radius." A closed Hamming ball changes only at

$$\mathcal{H}_n = \left\{0, \frac{1}{n}, \dots, \frac{n}{n}\right\}.$$

Consequently, a safe-radius set defined over all real $\delta \in [0,1]$ may be half-open and have no largest element. We report both the largest radius in $\mathcal{H}_n$ and the supremum over real radii.

4.2 An explicit smooth large-field family

For $u \in \mathbb{N}$, set

$$k_u = 64 \cdot 2^u = 2^{u+6}, \qquad n_u = 4k_u = 256 \cdot 2^u = 2^{u+8}, \qquad q_u = 3^{n_u}.$$

Let $\mathbb{F}_u = \mathbb{F}_{q_u}$. Since the multiplicative group of a finite field is cyclic of order $q_u - 1$, it suffices to check $n_u \mid (q_u - 1)$. Euler's theorem gives this directly:

$$\varphi(n_u) = \frac{n_u}{2}, \qquad \gcd(3, n_u) = 1, \qquad 3^{n_u/2} \equiv 1 \pmod{n_u}.$$

Squaring yields $3^{n_u} \equiv 1 \pmod{n_u}$. Thus $\mathbb{F}_u^\times$ contains a subgroup H_u of order n_u. We take $\mathcal{L}_u = H_u$ and

$$\mathcal{C}_u = \mathrm{RS}[\mathbb{F}_u, H_u, k_u].$$

The evaluation length and degree bound are powers of two, the domain is a multiplicative subgroup, and the rate is exactly $1/4$. This is the standard smooth-domain condition for FFT-based Reed–Solomon codes [ACFY24], and it is the explicit family used in the quantitative statement.

4.3 The field-size inequality

Fix $\varepsilon^* = 2^{-128}$. For the family above,

$$\binom{n_u}{k_u} \le 2^{n_u}.$$

Moreover n_u is even and $n_u/2 \ge 128$, so

$$\begin{aligned} q_u = 3^{n_u} &= (3^2)^{n_u/2} = 9^{n_u/2} \\ &\ge 8^{n_u/2} = 2^{3n_u/2} \ge 2^{n_u+128}. \end{aligned}$$

It follows that

$$\binom{n_u}{k_u} \le 2^{-128} q_u = \varepsilon^* |\mathbb{F}_u|.$$

The pair-MCA theorem uses the comparison

$$2^{128} \binom{n_u}{k_u + 1} \le q_u.$$

It follows from the same calculation, because $\binom{n_u}{k_u+1} \le 2^{n_u}$. Compared with the ordered certificate count, this removes a factor $k_u + 1$:

$$\binom{n_u}{k_u}(n_u - k_u) = (k_u + 1)\binom{n_u}{k_u + 1}.$$

The field is exponential in the block length. This is a valid explicit sufficiently-large-field regime, not a claim about practically sized fields or about every smooth code supplied to the Grand Challenge.

5 The Capacity-Grid List Theorem

The following argument applies to every injectively indexed Reed–Solomon evaluation domain; smoothness enters only when we instantiate the explicit family from Section 4.2.

Theorem 5.1 (Universal capacity-grid equality)**.** *Let $\mathcal{C} = \mathrm{RS}[\mathbb{F}, \mathcal{L}, k]$ have length n, where $1 \le k \le n$, and let $m \ge 1$. Then*

$$L_m\left(\mathcal{C}, \frac{n-k}{n}\right) = \binom{n}{k}.$$

In particular, the worst-case endpoint list size is independent of m.

Proof. We first prove the upper bound. Fix $w \in (\mathbb{F}^m)^n$. Every codeword in the ball agrees with w on at least k complete vector-symbol coordinates. Assign to each such codeword one k-subset of its agreement coordinates. This assignment is injective: if two interleaved codewords are assigned the same set S, then they agree with one another on S. In each component, two degree-$< k$ polynomials agree at the k distinct evaluation points indexed by S, and hence are equal. The two interleaved codewords are therefore equal. There are exactly $\binom{n}{k}$ possible sets S.

For the reverse inequality, choose one of the m components and define the received word

$$w_i^\star = (a_i^k, 0, \ldots, 0).$$

For $k < n$, the degree-k component is a classical RS deep-hole witness [KW16, Section 2]. The interpolation argument below counts its full interleaved endpoint list explicitly. For each k-subset $S \subseteq I$, put

$$V_S(X) = \prod_{i \in S} (X - a_i), \qquad f_S(X) = X^k - V_S(X).$$

The polynomial V_S is monic of degree k, so its leading term cancels that of X^k and $\deg f_S < k$. Use f_S in the distinguished component and zero polynomials in all other components to obtain an interleaved codeword c_S.

At evaluation coordinate i,

$$c_{S,i} = w_i^\star \quad \Longleftrightarrow \quad V_S(a_i) = 0 \quad \Longleftrightarrow \quad i \in S,$$

where the final equivalence uses injectivity of the evaluation points. Consequently c_S has distance exactly $n-k$ from $w^\star$, and its agreement set recovers S. The $\binom{n}{k}$ choices of S therefore give distinct codewords in the endpoint ball. This matches the upper bound. □

Theorem 5.2 (Failure on the next Hamming shell)**.** *Under the same hypotheses,*

$$\left| \Lambda \left(\mathcal{C}^{\equiv m}, 0, \frac{n-k+1}{n} \right) \right| \geq |\mathbb{F}|^m.$$

Proof. Fix $k-1$ evaluation points $S \subseteq \mathcal{L}$ and set

$$g_S(X) = \prod_{a \in S} (X - a).$$

For every $\lambda = (\lambda_1, \ldots, \lambda_m) \in \mathbb{F}^m$, the tuple

$$(\lambda_1 g_S, \ldots, \lambda_m g_S)$$

defines an interleaved codeword. These $|\mathbb{F}|^m$ codewords are distinct. The zero tuple has distance zero from the received word 0; every nonzero tuple vanishes on S and is nonzero at every evaluation point outside S, so its distance is exactly $n-k+1$. □

Theorem 5.3 (Fractional-target phase split)**.** *Fix $0 < \varepsilon^* < 1$, write $q = |\mathbb{F}|$, and call a radius δ safe when*

$$L_m(\mathcal{C}, \delta) \leq \varepsilon^* q.$$

For every $m \geq 1$, exactly one of the following regimes applies:

(i) *If $\varepsilon^* q < 1$, no radius in $[0,1]$ is safe.*

(ii) *If $1 \le \varepsilon^* q < \binom{n}{k}$, radius zero is safe but the capacity endpoint and every larger radius fail. The optimal sub-endpoint radius is not determined by the present argument.*

(iii) *If $\binom{n}{k} \le \varepsilon^* q$, the largest safe radius on $\mathcal{H}_n$ is*

$$\delta_{\mathrm{cap}} = \frac{n-k}{n} = 1 - \rho;$$

and over real radii the safe set is exactly

$$\left[0, \frac{n-k+1}{n}\right),$$

so it has no largest element and has supremum $(n-k+1)/n$.

Equivalently, the fractional target holds at the capacity endpoint if and only if

$$\binom{n}{k} \le \varepsilon^* |\mathbb{F}|.$$

Proof. Every codeword is contained in the radius-zero ball centered at itself, so no radius can be safe when $\varepsilon^* q < 1$. A radius-zero ball contains at most one codeword, proving its safety when $\varepsilon^* q \ge 1$.

Theorem 5.1 proves both the failure in the intermediate regime and the necessary-and-sufficient endpoint criterion. In the final regime, it and monotonicity of list balls prove safety through count radius $n-k$. A real radius strictly below $(n-k+1)/n$ defines the same ball as some count radius at most $n-k$. At the next grid point, Theorem 5.2 gives a list of size at least $|\mathbb{F}|^m \ge |\mathbb{F}| > \varepsilon^* |\mathbb{F}|$; monotonicity then rules out every larger radius. □

Corollary 5.4 (The explicit smooth family)**.** *For every $u \ge 0$, every $m \ge 1$, and $\mathcal{C}_u = \mathrm{RS}[\mathbb{F}_u, H_u, k_u]$ from Section 4.2, the conclusions of Theorem 5.3*(iii) *hold with $\varepsilon^* = 2^{-128}$. In particular, the largest safe Hamming-grid radius is* $3/4$.

Proof. Apply the inequality proved in Section 4.3. □

5.1 Scope relative to the Grand Challenge

Theorem 5.1 completely determines the endpoint list size for every supplied Reed–Solomon code, smooth or otherwise. Thus the displayed binomial field-size inequality is necessary and sufficient at that endpoint, not merely a convenient sufficient hypothesis. Corollary 5.4 supplies a concrete smooth family in the safe regime. The present argument does not determine the largest sub-endpoint radius for an arbitrary fixed smooth code in the intermediate regime

$$1 \le 2^{-128}|\mathbb{F}| < \binom{n}{k}.$$

The exponential alphabet is what places the explicit family in the final regime; it is not needed for the universal endpoint equality.

This distinction matters in light of known capacity results. Generic Reed–Solomon evaluation points can, for every fixed positive gap ϵ, achieve radius $1 - \rho - \epsilon$ over exponentially large fields with a list size bounded in terms of ϵ [BGM24]. This positive-gap statement differs from the exact endpoint equality above, and generic points do not supply the structured smooth domain required here. Conversely, near-capacity proximity-gap behavior can fail for particular smooth prime-field families [Kam26]. The theorem above imports neither result: it uses only polynomial interpolation and an explicit field-size comparison.

Recent work of Dao, Kominers, and Thaler [DKT26] gives quantitative list-decoding and MCA bounds on prescribed evaluation domains, including beyond-Johnson bounds under sufficiently large characteristic. Their positive agreement-gap counting results do not cover the exact agreement endpoint $A = k$; their beyond-Johnson counting hypotheses also exclude the characteristic-three family with $k \ge 64$ used here. Their Johnson-regime results in every characteristic remain a separate comparison. The present binomial bounds are not asserted to improve the sharper radius-dependent estimates in that literature, which also includes existing Lean/ArkLib formalizations.

6 Pair Mutual Correlated Agreement

6.1 Bad challenges

Let $p = (p_0, p_1) \in (\mathbb{F}^2)^I$, and write

$$y_\gamma = p_0 + \gamma p_1.$$

For $S \subseteq I$, say that S is δ-large when

$$\frac{|S|}{n} \geq 1 - \delta.$$

Define $\mathrm{Bad}_{\mathrm{MCA}}(\mathcal{C}, \delta, p)$ to be the set of $\gamma \in \mathbb{F}$ for which there exists a δ-large set S such that:

(i) y_γ agrees on S with a codeword of $\mathcal{C}$;

(ii) the pair p does not agree on the same S with a codeword of $\mathcal{C}^{\equiv 2}$.

The MCA error is

$$\varepsilon_{\mathrm{MCA}}(\mathcal{C}, \delta) = \max_{p \in (\mathbb{F}^2)^I} \frac{|\,\mathrm{Bad}_{\mathrm{MCA}}(\mathcal{C}, \delta, p)|}{|\mathbb{F}|}.$$

This is the two-word same-set event appearing in [BSCH+26, Theorem 4.6, full version]. That source uses polynomials of degree at most its parameter, so its parameter is $k - 1$ in the degree-$< k$ convention used here. Our bound below is uniform in the radius; the cited literature gives sharper radius-dependent bounds in its stated regimes. At fixed rate and a fixed positive gap below the Johnson radius, the cited theorem gives an exceptional-challenge count linear in n, generally much smaller than the binomial bound below. Broader polynomial-generator formulations of MCA for linear codes are studied by Bordage et al. [BCGM26].

Theorem 6.1 (Uniform pair-MCA certificate bound)**.** *Let $\mathcal{C} = \mathrm{RS}[\mathbb{F}, \mathcal{L}, k]$ have injectively indexed evaluation domain of length n, where $1 \leq k \leq n$. For every $\delta \in \mathbb{R}$ and every $p \in (\mathbb{F}^2)^n$,*

$$|\,\mathrm{Bad}_{\mathrm{MCA}}(\mathcal{C}, \delta, p)| \leq \binom{n}{k+1}.$$

Consequently,

$$\varepsilon_{\mathrm{MCA}}(\mathcal{C}, \delta) \leq \frac{\binom{n}{k+1}}{|\mathbb{F}|}.$$

Smoothness is not required for this bound.

Proof. Fix a bad challenge γ, and choose a witnessing set S. Necessarily $|S| > k$. Indeed, when $|S| \leq k$, interpolate each component of p on S by a polynomial of degree $< k$; the resulting pair codeword agrees with p on S, contradicting condition (ii).

Choose a k-subset $T \subset S$. Let a_T, b_T be the unique degree-$< k$ polynomials interpolating p_0, p_1, respectively, on T. Condition (ii) says that the pair interpolant (a_T, b_T) fails somewhere on S. Choose such an $i \in S \setminus T$, and set $U = T \cup \{i\}$. Condition (i) remains true after restriction from S to U. The pair still cannot agree with $\mathcal{C}^{\equiv 2}$ on U: any purported pair codeword would agree with (a_T, b_T) on T, hence equal it by k-point determination, contradicting the mismatch at i. Thus U is a $(k+1)$-coordinate obstruction for γ.

A fixed U obstructs at most one challenge. Suppose distinct γ, η were both bad on U, with scalar codewords c_γ, c_η agreeing with $p_0 + \gamma p_1, p_0 + \eta p_1$, respectively. By linearity,

$$d = \frac{c_\gamma - c_\eta}{\gamma - \eta}, \qquad a = c_\gamma - \gamma d$$

are codewords. On U, they agree with p_1 and p_0, respectively, so (a, d) is a pair codeword agreeing with p on U, a contradiction.

Make the finite choices deterministic. Mapping each bad challenge to its unordered obstruction U is injective, and there are $\binom{n}{k+1}$ such subsets. Dividing by $|\mathbb{F}|$ proves the error bound. Notice that the large-set inequality is no longer used after the witness is extracted; this is why the bound is uniform in δ. □

Corollary 6.2 (Explicit smooth family at 2^{-128})**.** *For every member $\mathcal{C}_u$ of the family in Section 4.2, every $\delta \in [0, 1]$, and every pair word p,*

$$\frac{|\operatorname{Bad}_{\mathrm{MCA}}(\mathcal{C}_u, \delta, p)|}{|\mathbb{F}_u|} \le 2^{-128}.$$

Thus, for this family and this same-set pair-MCA definition, every normalized radius is safe and the largest admissible radius is 1*.*

Proof. Section 4.3 proves

$$2^{128} \binom{n_u}{k_u + 1} \le q_u.$$

Apply Theorem 6.1. □

6.2 An exact small-radius boundary instance

The uniform certificate bound is sufficient but need not locate the first unsafe radius. Write

$$B_{\mathcal{C}}(r) = \max_p |\operatorname{Bad}_{\mathrm{MCA}}(\mathcal{C}, r/n, p)|$$

for the exact worst bad-challenge count at count radius r. Consider

$$q = 2^{128} + 192513 = 340282366920938463463374607431768403969.$$

The accompanying Lean development certifies that q is prime and $q \equiv 1 \pmod{4096}$. Hence $\mathbb{F}_q^\times$ contains an order-4096 subgroup, which we use as the evaluation domain for the smooth RS code with $n = 4096$ and $k = 1024$.

Theorem 6.3 (Exact 128-bit small-radius transition)**.** *For this code,*

$$B_{\mathcal{C}}(0) = 1, \qquad B_{\mathcal{C}}(1) \geq 2.$$

Consequently, for the 2^{-128} MCA error target, its greatest safe count radius is 0 and its first unsafe count radius is 1. For the relative-radius definition over all $\delta \in [0, 1]$, the safe set is

$$\left[0, \frac{1}{4096}\right),$$

so the real-radius supremum is $1/4096$ and is not attained.

Proof. At count radius zero, every large agreement set is the whole coordinate set. A fixed set supports at most one bad challenge, while a unit-spike affine line realizes one; hence $B_{\mathcal{C}}(0) = 1$. At count radius one, take two coordinates outside a k-point interpolation set. The affine line from the unit spike at the first coordinate to the unit spike at the second has distinct bad challenges 0 and 1, proving $B_{\mathcal{C}}(1) \geq 2$.

The certified field lies in the exact budget window

$$2^{128} \leq q < 2^{128} \cdot 2.$$

Thus the integer criterion $2^{128} B_{\mathcal{C}}(r) \leq q$ holds at $r = 0$ and fails at $r = 1$. Monotonicity of the bad-count spectrum gives the two grid endpoints.

Finally, if $0 \leq \delta < 1/n$, the inequality $|S|/n \geq 1 - \delta > 1 - 1/n$ forces the integer $|S|$ to equal n. The bad sets are therefore exactly those at count radius zero throughout that half-open interval. At $\delta = 1/n$, sets missing one coordinate become admissible, and the two-spike line already violates the budget. Monotonicity preserves failure for larger radii, proving the real-radius claim. □

6.3 Distance interpretation

Let

$$D_2(p) = \Delta(p, \mathcal{C}^{=2}), \qquad D_1(y) = \Delta(y, \mathcal{C}).$$

For a challenge outside the MCA bad set, scalar closeness on a large agreement set forces pair closeness on that same set. For $\delta \in [0, 1]$, this gives the distance implication

$$D_2(p) \geq \delta \Longrightarrow D_1(y_\gamma) \geq \delta$$

for every MCA-good γ, with the fraction of excluded challenges bounded by $\varepsilon_{\mathrm{MCA}}(\mathcal{C}, \delta)$.

To see the boundary convention explicitly, suppose that $D_1(y_\gamma) < \delta$, and choose a closest scalar codeword c. Let $S = \{i : y_\gamma(i) = c(i)\}$. Then $|S|/n = 1 - D_1(y_\gamma) > 1 - \delta$, so S is δ-large. Since γ is MCA-good, a pair codeword agrees with p on S. Hence $D_2(p) \leq D_1(y_\gamma) < \delta$. Taking the contrapositive proves the displayed implication, including its non-strict $\geq$ boundary.

This is a challenge-indexed statement. Replacing the sampled challenge by one fixed field element would require a separate theorem that the selected element is good for every pair, which does not follow from a fractional MCA bound.

6.4 Endpoint interpretation

At $\delta_{\mathrm{cap}} = 1 - k/n$, interpolation on any k coordinates shows that

$$D_1(y) \leq \delta_{\mathrm{cap}} \quad \text{and} \quad D_2(p) \leq \delta_{\mathrm{cap}}$$

for every scalar word y and pair word p. Moreover, on every set S of size at most k, each component of p extends to a degree-$< k$ polynomial. Such an S can therefore never witness an MCA failure. Every bad endpoint challenge must instead have a witness with $|S| \geq k + 1$.

Consequently, the endpoint distance implication above is an equality non-collapse statement: when $D_2(p) = \delta_{\mathrm{cap}}$, a good challenge must not lower the projected distance strictly below δ_{cap}. Theorem 6.1 supplies a uniform quantitative bound on the excluded challenges.

6.5 Commit-then-challenge order

The mathematical experiment fixes p before sampling $\gamma \leftarrow \mathbb{F}$. This ordering prevents the pair from being selected as a function of the challenge. A

full protocol implementation must bind p before deriving γ, but the coding theorem needs only the resulting quantification and probability space; it does not depend on a complete blockchain or commitment protocol.

6.6 Scope

The theorem is universal over pair words and injective RS evaluation domains, but its 2^{-128} corollary uses the explicit exponential-size field. For a different supplied field, the same certificate proof gives the sufficient comparison

$$\binom{n}{k+1} \leq 2^{-128}|\mathbb{F}|.$$

Relative to the older ordered-certificate count used in this development, the bound saves the exact factor $k+1$, since $\binom{n}{k}(n-k) = (k+1)\binom{n}{k+1}$. This is not a uniform improvement over the cited literature bounds. If that inequality fails, this proof does not determine the optimal radius or show that the MCA target is impossible; it only ceases to certify the requested error.

For an arbitrary supplied finite code, radius r/n meets the 2^{-128} target exactly when

$$2^{128} B_{\mathcal{C}}(r) \leq |\mathbb{F}|.$$

This is also the exact formal equivalence: the Lean theorem identifies the real-valued operational error predicate at normalized radius r/n with the displayed integer inequality, so no rounding relaxation is hidden here. Thus a uniform upper-bound certificate at r, together with one concrete affine line exceeding the budget at $r + 1$, pins both the largest safe and first unsafe Hamming-grid radii. The remaining middle regime is the code-specific problem of proving those matching bounds; smoothness by itself does not determine this incidence spectrum.

The result is for the two-coordinate, same-set MCA experiment defined above. Broader notions or higher-dimensional challenge families require separate statements.

As a separate concrete/application result, the repository also checks the canonical runtime field $q = 12289$ with $n = 4096$ and $k = 1024$. Since $q < 2^{128}$ while its radius-zero unit-spike line has one bad challenge, that concrete code has no safe count radius for the 2^{-128} MCA error target. This runtime-facing fact is not used by the standalone large-field or boundary-field theorems above.

7 Application Interface

Paper 2 specifies the shared cochain construction, snapshot/decoder context, coordinate layout and public observation profile [vdV26b]. This section recalls only the data needed to state the projection and proves its quantitative validity, distance and challenge comparisons. The commitment-security, protocol gate/evidence and state-transition contracts remain with that construction. Paper 1 supplies the HCP and cellular Hodge primitives [vdV26a]; the exact-component calibration obstruction is proved in Proposition 7.3 below.

The coding theorem is stated over the complete received-word space and does not depend on a particular source of observations. An application may nevertheless inherit the result through a distance-aware map.

Let O be a type of raw presented observations, let $G \subseteq O$ be the exact/valid locus, and let

$$\pi : O \longrightarrow (\mathbb{F}^2)^I$$

be the extracted pair word. A source distance $\rho_O : O \to [0, \infty)$ is meaningful only when O includes arbitrary or noisy presentations; if every element of O already contains a validity witness, then the proximity problem may collapse to the exact code locus.

The Lean companion packages this interface as a source validity model and a source-to-code calibration. Its only quantitative field is the pointwise inequality

$$\rho_O(o) \leq D_2(\pi(o)).$$

It yields, uniformly in δ,

$$\rho_O(o) \geq \delta \Longrightarrow D_2(\pi(o)) \geq \delta.$$

Under this comparison, the proved universal list theorem applies to $\pi(o)$; the proved pair-MCA theorem applies through the same interface once the application fixes its pair before sampling the challenge.

There are two standard ways to discharge the comparison. Defining $\rho_O = D_2 \circ \pi$ makes it reflexive. A pre-existing source metric requires a mathematical proof of $\rho_O \leq D_2 \circ \pi$. Exact witnesses may certify membership in G, but the carrier O must still include arbitrary received observations.

Protocol-specific observation formats, commitments, and runtime verification are intentionally outside the proof of Theorems 5.1, 5.2 and 6.1. Such developments may cite the two coding theorems through the interface above; neither theorem assumes that every received word comes from an application-specific witness space.

7.1 Projective coordinates in the projection chain

Let $V = \mathbb{R}^n$ have its fixed Euclidean inner product, and let $w_1, \dots, w_r$ be an independent frame spanning $W \le V$. Its exterior coordinate is

$$c = w_1 \wedge \cdots \wedge w_r \in \bigwedge^r V.$$

An independent frame has nonzero exterior product. A different basis of the same subspace changes c by the nonzero determinant of its change-of-basis matrix. The projective class therefore depends only on W:

$$c \sim c' \quad \Longleftrightarrow \quad \exists u \in \mathbb{R}^\times,\ c' = uc.$$

The raw carrier is the scalar-orbit quotient of the whole exterior power, including its zero orbit. The conclusions below use the nonzero decomposable orbits determined by genuine frames.

A related specialized reconstruction uses restricted Plücker coordinates for maximal isotropic subspaces [DK26]. The argument here uses the full projective point of an arbitrary real subspace at the fixed ambient metric.

Proposition 7.1 (Projective Plücker invariance)**.** *Two independent frames spanning the same finite-dimensional subspace have equal projective exterior coordinates.*

Proof. Both exterior products are nonzero elements of the one-dimensional top exterior power of that subspace. They differ by a nonzero scalar and hence have the same orbit. □

Recovery from the full point.

Proposition 7.2 (Recovery from the full projective point)**.** *Let $w_1, \dots, w_r$ be an independent frame for $W \subseteq \mathbb{R}^n$, and put $c = w_1 \wedge \cdots \wedge w_r$. Then*

$$W = \{x \in \mathbb{R}^n : x \wedge c = 0\}.$$

This reconstruction depends only on $[c]$. Consequently, two genuine rank-r frame witnesses have the same full projective point if and only if they present the same subspace. At the fixed ambient inner product, the full point determines P_W and $I - P_W$.

Proof. If $x \in W$, the frame with x adjoined is dependent and its exterior product vanishes. If $x \notin W$, the adjoined frame is independent and its exterior product is nonzero. Multiplying c by a nonzero scalar does not change this annihilator. The projector is the unique orthogonal projection onto the reconstructed subspace. □

The zero orbit has the whole ambient space as its annihilator, which is why the statement requires a genuine frame. For the operative harmonic subspace $W = \ker \Delta_x$, the reconstructed projector and b give

$$h = P_W b.$$

The positive spectrum of Δ_x remains additional operator data. Replacing the full real coordinates by encoded data or a digest requires a separate reconstruction procedure.

The geometric source can enter the existing extraction chain through

$$[w_1 \wedge \cdots \wedge w_r] \xrightarrow{\mathrm{enc}} c \xrightarrow{E} p_c \xrightarrow{\bmod q} \bar{p}_c \xrightarrow{\gamma} y_\gamma.$$

Here enc is the supplied projective-to-cochain map, c is the original commitment payload, and the remaining maps are the actual plane extraction, coefficient reduction and challenge evaluation specified below. The first map requires its own realization and validity hypotheses; reconstruction of a subspace does not construct that codec. The projection-chain theorem composes the supplied validity and near-distance comparisons by applying each stage in order and using transitivity of the distance bounds. A source-far conclusion requires a separate comparison in the direction used by the application calibration; it is not manufactured by composing these near-side maps. Full-coordinate reconstruction supplies no such quantitative inequality.

7.2 The existing committed-cochain projection

The project-specific application uses Paper 2's existing commitment payload [vdV26b],

$$c = (q, N, \mathbf{c}, \tau).$$

Here $\mathbf{c}$ is the serialized integer edge-value list. The Lean carrier `CommittedCochain` is an abbreviation for the same `CochainCommitmentData` used by the commitment and artifact layers. A selected observation exposes that payload; it does not replace it with a second commitment object. The construction and authentication conditions of that observation belong to Paper 2; here the payload is the input to the actual extraction map.

Fix an extraction context consisting of an expected edge count, selected indices $\iota(i, j)$, and a coefficient interpretation $a : \mathbb{Z} \to \mathbb{Z}$. For an aligned payload, the actual plane projection is

$$p_c(i, j) = a(\mathbf{c}_{\iota(i,j)}), \qquad j \in \{0, 1\}.$$

Alignment ensures that every selected index is in bounds. The public plane-view certificate records equality with this extraction from the observed payload. The canonical instantiation reduces both coordinates modulo $q = 12289$, with RS length $n = 4096$ and degree bound $k = 1024$:

$$\bar{p}_c(i,j) = p_c(i,j) \bmod q, \qquad y_\gamma(i) = \bar{p}_c(i,0) + \gamma \bar{p}_c(i,1).$$

Thus one observed commitment supplies the full pair before the challenge is evaluated. The `PublicCalibration` module implements this extraction and proves its equality with the observed-payload projection.

On integer planes whose coordinates lie in $[0, q)$, coordinatewise reduction is a bijection onto the field-pair space. It preserves whole-pair Hamming distance. Let $\mathcal{C}_A$ be the pair code pulled back to this canonical integer alphabet, and let D_A be relative Hamming distance to $\mathcal{C}_A$. The existing Lean development proves

$$D_A(p_c) = D_2(\bar{p}_c).$$

The public-observation calibration uses $D_A(p_c)$ as its source score, so its all-radius farness implication is already proved. Its observation carrier requires aligned, canonical-region payload views without requiring pair-codeword membership.

An application retaining an independently chosen cochain, Hodge, or Plücker disagreement ρ_c still needs a quantitative comparison, for example

$$\rho_c(c) \geq r_S \quad \Longrightarrow \quad D_A(p_c) \geq r_T.$$

The equality above does not prove this upstream comparison. In particular, changing unselected edge values leaves p_c unchanged; any source metric sensitive to those values needs an admissible-family restriction or a separate control theorem.

The storage interface now fixes the same snapshot, payload interpretation and public artifact in one context, while keeping exact acceptance separate from raw context binding. For its decoded cochain, one independent geometric score is

$$\rho_{\mathrm{ex}}(c) = \| P_{(\operatorname{im}\delta_0)^\perp} \operatorname{dec}_S(c) \|.$$

It vanishes exactly on accepted exact payloads once the context and shape conditions hold. A sufficient comparison criterion is an exact comparison cochain $\lambda(z) \in \operatorname{im}\delta_0$ for every target pair-codeword z, together with a constant $A > 0$ and the bound

$$\| \operatorname{dec}_S(c) - \lambda(z) \| \leq A\, \delta_H(z, \bar{p}_c).$$

Choosing an attained nearest codeword and applying orthogonal-projection contractivity gives

$$\rho_{\mathrm{ex}}(c) \le A D_2(\bar{p}_c), \qquad \rho_{\mathrm{ex}}(c) \ge r_S \Longrightarrow D_2(\bar{p}_c) \ge r_S/A.$$

The storage calibration module proves this implication from the lift and decoder bounds. It does not construct those application-specific inputs. The canonical storage adapter identifies the real-infimum pair distance with the attained generic minimum and joins both comparisons to the actual observed artifact.

7.3 Source-metric obstructions and a no-wrap profile

Fix a real cellular cochain complex $C^0 \xrightarrow{\delta_0} C^1 \xrightarrow{\delta_1} C^2$ with its chosen inner products and harmonic subspace $\mathcal{H} = \ker \Delta_1$. The full Hodge sample $b = h + \delta_0 a + \delta_1^* v$ and its exact-component view $\delta_0 a$ retain different data.

Proposition 7.3 (Exact-component obstruction)**.** *Suppose $v \in C^2$ and $\delta_1^* v \neq 0$. On the unrestricted Hodge-noise carrier, the samples*

$$b_t = t\delta_1^* v, \qquad t \in \mathbb{R},$$

have zero harmonic message and zero exact component, while

$$\|P_{\mathcal{H}^\perp} b_t\| = |t| \, \|\delta_1^* v\|.$$

Thus this full-observation residual does not factor through the exact component. If an extraction depends only on that component and its zero image has target distance zero, it cannot preserve any positive target far radius uniformly on this unrestricted carrier, at any source radius.

Proof. A coexact vector is orthogonal to $\mathcal{H}$, so its orthogonal projection onto $\mathcal{H}^\perp$ is itself. Homogeneity gives the displayed norm. The exact-component view is constantly zero, whereas this norm is zero at $t = 0$ and positive at $t = 1$, proving nonfactorization. For any fixed source radius, choose positive t large enough to exceed it. The extracted value still has distance zero, contradicting every positive target far radius. □

A restricted noise or realization profile may exclude this family, so a comparison for such a profile requires its explicit hypotheses. The scores

$$\|P_{H^\perp} b\|, \qquad \operatorname{dist}(b, \operatorname{im} \delta_0), \qquad \text{a modular cochain or code distance}$$

measure harmonic residual, distance to exactness, and modular or code disagreement, respectively. A modular distance needs its own carrier and normalization.

The canonical systematic profile below has no faces and satisfies $\delta_1 = 0$, so the proposition's nonzero-coexact premise is unavailable in that profile.

Canonical representatives alone do not make a finite-field RS code into a proper real-linear exact-cochain locus. Suppose q is prime, the evaluation points are distinct and $k \geq 2$. For a scalar field word w, write $\ell(w) \in \mathbb{R}^I$ for its canonical coordinate lift. The lifts of the affine RS words already span $\mathbb{R}^I$. Indeed, for each coordinate i, put $v_j = a_j - a_i$. Canonical representatives satisfy

$$\mathbf{1} - \ell(v) + \ell(v - \mathbf{1}) = qe_i.$$

Only the i-th coordinate wraps at zero. Both affine words and the constant word belong to RS, so every coordinate vector belongs to the real span of the lifted codewords. Applying this argument separately to both columns proves the corresponding assertion in $\mathbb{R}^{I \times \{0,1\}}$.

Consequently, if a real-linear decoder T has any nonexact output, there is no constant A for which

$$\operatorname{dist}(T\ell(w), \operatorname{im} \delta_0) \leq AD_2(w) \quad \text{for every field-pair word } w.$$

Such a bound would make all lifted pair-codewords exact, hence make the decoder's entire image exact. This obstruction concerns a global comparison with no additive loss. It leaves restricted profiles, additive defects and separately justified threshold implications open.

One explicit restricted profile retains the original Euclidean source metric. Choose an information set S of size k and let $A_{i,s} \in \mathbb{F}_q$ be the actual systematic RS interpolation coefficients. Lift them to integers $\widetilde{A}_{i,s} \in [0, q)$, and define the real generator and integer check by

$$(G_{\mathbb{R}} u)_i = \sum_{s \in S} \widetilde{A}_{i,s} u_s, \qquad (Hz)_i = z_i - \sum_{s \in S} \widetilde{A}_{i,s} z_s.$$

In Paper 2's systematic construction, both columns use the complete coordinate layout with identity coefficient map. Its two-vertex, one-edge cellular sheaf realizes $\operatorname{im} \delta_0 = \operatorname{im}(G_{\mathbb{R}} \oplus G_{\mathbb{R}})$; this equality is proved from its restriction maps. The decoder reads the same original integer payload into its real edge coordinates.

Assume $|z_{i,j}| \leq B$, $B > 0$, and the integer-syndrome condition $|(Hz_{\cdot,j})_i| < q$ for all i, j. If the reduced pair is a codeword, every integer check is divisible by q; the syndrome bound forces it to vanish. Thus codeword validity reflects

geometric exactness. If the source is nonexact, its extracted pair has distance at least $1/n$. Since $\rho_{\mathrm{ex}}(z) \le \|z\| \le B\sqrt{2n}$, these facts give

$$\rho_{\mathrm{ex}}(z) \le Bn\sqrt{2n}\, D_2(\bar{z}), \qquad \rho_{\mathrm{ex}}(z) \ge r \Longrightarrow D_2(\bar{z}) \ge \frac{r}{Bn\sqrt{2n}}.$$

The source score remains Euclidean distance to exact cochains. The no-wrap condition bounds the integer syndrome, not just individual representatives. An outside-information-coordinate unit error in one column has nonzero real syndrome, satisfies this profile and gives a positive source distance.

This construction is a coarse positive-radius example. For a nonempty source far region its certified target radius is at most $1/n$; it does not supply a capacity- or Johnson-scale geometric application theorem. Positive decoded farness concerns raw received payloads and excludes already accepted exact anchors. The constructed systematic sheaf is an explicit application model, not an automatic identification with an arbitrary HCP presentation, and no encryption-security claim follows from this construction.

7.4 Three native metrics for modular cochains

The same shared cellular construction has a global comparison in its native modular metrics. The generic modular result uses a prime field $\mathbb{F}_q$, an injective RS domain with $0 < k \le n$, and a chosen information set of size k; subgroup smoothness is not required. Recall that the shared cochain valid locus is specified intrinsically by an actual integral opening and integer carry:

$$\delta_0 a = z + qm.$$

This is modular exactness, distinct from real decoded exactness. Reducing the actual integer interpolation coefficients proves that modular exactness is equivalent to the original RS pair-code predicate. No no-wrap condition is needed for this modular statement.

Let $d_C(z)$ be scalar-coordinate edit distance to these modularly exact cochains, normalized by $2n$. Let $d_P(z)$ be whole-pair edit distance to modularly exact integer planes, normalized by n. The RS distance is $D_2(\bar{z})$. For two words, their mismatch counts satisfy $h_P \le h_C \le 2h_P$. Taking attained nearest codewords gives

$$d_C(z) \le d_P(z) \le 2d_C(z).$$

The plane/RS distance equality is also a theorem. Reduction cannot create a mismatch. Conversely, lift a field codeword pair by retaining each already

matching received integer pair, and using canonical representatives at the remaining positions. The lifted word is modularly exact and has exactly the field word's mismatched positions. Thus

$$\boxed{d_C(z) \le d_P(z) = D_2(\bar{z}) \le 2d_C(z)}.$$

The scalar representative-lifting argument also identifies d_C with the corresponding flattened finite-field code distance. These independently defined native distances are compared through the actual maps. A unit-scale near contract must use the explicitly scaled local score $2d_C$; the far implication uses $d_C \le D_2$.

The factor two is sharp for proper RS codes. Put $u_i = a_i^k$, and take the pair $(u, 0)$. Its whole-pair distance is $(n-k)/n$, while its scalar-coordinate distance, normalized by $2n$, is $(n-k)/(2n)$. The same representative-lifting equalities transfer this witness to the native integer metrics, so the upper bound cannot be replaced by a unit-factor near comparison.

There is a nonempty capacity-scale family in these modular metrics. For $0 < k < n$, take $u_i = a_i^k$ and both pair columns equal to u. A polynomial of degree below k agrees with u on at most k positions, while interpolation supplies a codeword agreeing on any chosen k positions. Applying this separately to both columns, and to their common interpolation, proves

$$d_C(z) = d_P(z) = D_2(\bar{z}) = \frac{n-k}{n}.$$

For the canonical instance $(q, n, k) = (12289, 4096, 1024)$, representatives of these two columns serialize into the original commitment payload in the declared complete layout, and all three distances are $3/4$. The payload is a well-formed raw proximity input. Under the matching decoder, accepted exact anchors have zero native distances. This is a native modular source statement, not a capacity-scale Euclidean Hodge-distance theorem.

For this diagonal pair, the challenged word is $(1+\gamma)u$. Every $\gamma \neq -1$ preserves the scalar capacity distance and is MCA-good; $\gamma = -1$ collapses to zero. At every nonnegative tested radius its bad set is exactly $\{-1\}$, with uniform fraction $1/q$. The existing canonical mix is noncancelling because its order is $4096 > 2$.

The restriction to the stated line profile is essential. For an arbitrary fixed nonzero γ_0, the pair $(u, -\gamma_0^{-1}u)$ has scalar-coordinate and pair distance $(n-k)/n$, but its challenge at γ_0 is zero. A global fixed-mix far-preservation claim therefore fails even on capacity-far pairs. The general protocol must bind the pair before sampling its challenge or prove goodness for its actual restricted line profile.

For the full HCP residual, Proposition 7.3 retains the nonzero-coexact and unrestricted-source hypotheses. An exact-component-only view needs a justified quotient or restricted source profile before it can support a full-HCP far-region claim. Applying the proposition to a serializer also requires proving that its extraction factors through that decoded view.

For scalar transfer, the pair must remain fixed before a random challenge is sampled, or a chosen fixed challenge must be proved MCA-good. The canonical fixed mix $g = 1331$ is a different assertion from a bound on the fraction of bad random challenges. Restricting integer representatives makes the pair faithful; it does not make one scalar evaluation injective.

The current canonical field is also distinct from the sufficiently large family of Section 4. For the full canonical pair space, the existing small-radius certificate already exceeds the 2^{-128} budget at count radius zero because $12289 < 2^{128}$. This does not determine the spectrum on a restricted commitment image. Adopting $\mathbb{F}_{3^{n_u}}$ would require its own field-element encoding: ordinary integer casts reach only the prime subfield, and the current wire-modulus convention is unchanged.

7.5 Scope of the instantiated application

Paper 2's constructed systematic profile fixes the concrete cellular model, full layout, identity integer coefficient map, raw payload carrier and intrinsic modular valid locus. The results proved here compare its three independently defined native metrics, construct a nonempty capacity-far original-payload family and analyze its challenges. The calibration is packaged through the existing source-to-source and source-to-code interfaces; the existing near projection uses the declared local scale $2d_C$. The public-observation adapter retains the original artifact and its actual extracted pair.

An application to a different maintained sheaf must prove the corresponding source and coefficient identifications. An independently chosen HCP/frame score needs its own upstream comparison; the native modular theorem does not identify that score with edit distance. The global canonical-lift and fixed-challenge obstructions constrain such stronger claims. The restricted Euclidean theorem and the diagonal fixed-mix example retain their stated profiles.

The selected profile gives a concrete relation for comparison with existing commitment interfaces. In its complete identity layout, reduce the original integer payload to $v \in \mathbb{F}_q^{2n}$. Modular validity is the actual coboundary-image relation, proved equivalent to paired RS membership. For a fixed challenge, $v \mapsto v_0 + \gamma v_1$ is $\mathbb{F}_q$-linear. This identification uses the chosen

identity coefficient map; the earlier generic interpretation $a : \mathbb{Z} \to \mathbb{Z}$ is arbitrary.

Position-binding vector commitments [CF11] and linear-map vector commitments [CNR+22] concern authenticated coordinate or function outputs. That authentication does not itself give the native source-distance comparison above. A comparison with WHIR's constrained-RS proximity interface [ACFY24] additionally identifies the polynomial constraints and fixed-pair challenge experiment. The fixed-mix counterfamily prevents replacing this analysis by an arbitrary scalar fold. Commitment/opening/verifier algorithms, metadata binding and their security games belong to Paper 2. This paper proves the stated relation and metric comparisons, without a new binding/hiding primitive, priority claim or verification-cost advantage. These mathematical results do not require completion of Paper 2's ledger, consensus protocol or Rust implementation.

8 Formalization and Proof Ledger

8.1 Standalone hierarchy

The machine-checked development separates generic coding mathematics from application bridges:

`Proximity/Core/BlockCode`
: Relative distance, balls, and universal list-bound predicates.

`Proximity/Core/ReedSolomon`
: Evaluation-domain definitions, RS codes, and interleaving.

`Proximity/Core/PlanePair`
: Plane–pair–line equivalences.

`Proximity/Core/FiniteThreshold`
: Safe finite grids, greatest-safe and first-unsafe values, and adjacent safe/unsafe threshold certificates.

`Proximity/Core/Distance` and `Proximity/Core/RadiusShell`
: Attained minimum distances, closed and strict nearness, and the exact half-open real intervals between consecutive Hamming-count shells.

`Proximity/Capacity/Parameters`
: Explicit ε^* and field-size inequalities.

`Proximity/Capacity/Realization` and `Proximity/Capacity/ListPhase`
: Actual Galois fields, exact-order subgroup domains for every family member, and instantiated list/MCA grid and real-radius conclusions.

`Proximity/ListDecoding/Capacity`
: Agreement-set injection, the matching received-word construction, the universal endpoint equality, the fractional-target equivalence, and the next-shell coefficient-vector injection.

`Proximity/ListDecoding/PhaseSplit`
: All three fractional-target regimes, the exact endpoint-safe real interval, its excluded supremum, and the largest safe grid count.

`Proximity/MCA/Definitions`
: Arbitrary-block-code same-set MCA definitions at count and relative radii.

`Proximity/MCA/Threshold`
: Exact worst bad-count spectrum and the equivalence between operational real error at r/n and the integer $2^{-\lambda}$ budget.

`Proximity/MCA/SpectrumCertificate`
: Supplied-code grid-threshold pinning from a uniform upper bound at one radius and a concrete over-budget affine line at the next radius.

`Proximity/MCA/SmallRadius`
: Universal RS unit-spike and two-spike certificates, including $B_{\mathcal{C}}(0) = 1$ for every proper RS code.

`Proximity/MCA/BoundaryField`
: Primality, order-4096 smooth domain, and the exact 128-bit radius transition for $q = 2^{128} + 192513$.

`Proximity/MCA/RealThreshold`
: Count/real same-set event correspondence and the boundary-field safe interval $[0, 1/4096)$, including supremum and nonattainment.

`Proximity/MCA/Pair`
: RS compressed interpolation certificates, bad-challenge injection, the uniform cardinality bound, and the explicit-family error corollary.

`Proximity/Application/Calibration`
: Generic source-validity model, pointwise distance non-collapse, and the uniform source-far to target-far implication.

`Proximity/Application/DistanceCalibration`
: Identification with attained scalar/pair code distance, interpolation covering bounds, and good-challenge farness with its exact boundary.

`Proximity/Core/CanonicalLift` and `Proximity/Core/Systematic`
: Canonical real-lift spanning, actual systematic interpolation and integer/real syndrome compatibility. These are proved algebraic facts.

`Proximity/Core/InterleavedMetric`
: Native scalar-coordinate and whole-pair Hamming comparisons, retaining their different normalizations.

`Proximity/Core/MetricSharpness`
: The degree-k first column with zero second column attains factor two between pair and scalar-coordinate distance for proper RS codes.

`Proximity/Core/DeepHole` and `Proximity/Core/FixedChallenge`
: Exact scalar/pair/flattened capacity distances for monomial received words, and capacity-far counterexamples to arbitrary fixed-challenge compression.

`Proximity/MCA/ZeroDirection` and `Proximity/MCA/Diagonal`
: Explicit all-good and single-cancelling-challenge profiles, with their exact same-set bad events.

The standalone import closure does not depend on HCP, protocol, consensus, or runtime axiom bundles. Both directions of the capacity-grid equality and its exact fractional-target criterion are theorems, not axioms. The next-shell lower bound is also machine-checked. The pair-MCA theorem and its 2^{-128} family instantiation are machine-checked as well. The generic family corollaries retain their parameterized forms. The realization module additionally constructs $\mathbb{F}_{3^{n_u}}$, an exact-order n_u root, and its injectively indexed subgroup for every u, and instantiates those corollaries without a realization hypothesis. All source modules in this ledger live under `lean/Proximity`; `lean/ProximityCoding.lean` is only the standalone aggregate import target, not a second source hierarchy. The detailed object dictionary, theorem correspondence, construction ownership and remaining mathematical extensions are recorded in `papers/paper3/SEMANTIC_PARITY.md`.

8.2 Auxiliary source and projective proofs

The optional geometric source in Section 7.1 uses the existing `Proximity/Concrete/GrassmannianPlucker/Reconstruction` module. Its `recoverSubspace_frame` and `recoverHarmonicProjector_frame` theorems recover the full subspace and its fixed-metric projector. `TCB/PluckerObservation.recover_eq_operativeHarmonicComponent` adds the explicit operative-kernel identification. `Proximity/Concrete/ProjectionChain/Core` composes the supplied projective, cochain, plane and RS maps and their validity/near comparisons. The exact-component obstruction in Proposition 7.3 uses the existing `Sheaf/ObservationQuotient` proofs `HodgeEncryptedCochain.not_residual_norm_factors_exactComponent` and `HodgeEncryptedCochain.not_far_preserving_exactComponent`. They retain the unrestricted Hodge carrier, a nonzero coexact direction, exact-component factorization and zero target distance at the zero image. The canonical systematic sheaf has $\delta_1 = 0$, so this nonzero-coexact premise is unavailable there.

These source-obstruction and projective results are canonical auxiliary proofs outside the configured 52-target coding and 22-target native-modular application selections. They do not enter the standalone coding import closure; moving a statement into this manuscript does not add a selected target. Their formal owners and the independent source metrics remain distinct from Paper 2's codec, visibility and authentication contracts. The projection-chain composition gives validity and near-side handoffs; source-far preservation requires the separately stated calibration.

8.3 Status ledger

Obligation	Status	Required endpoint
Count/relative ball correspondence	Lean-proved	Equality at r/n, for $n > 0$
Plane–pair equivalence	Lean-proved	Bijection and Hamming isometry
Plane–affine-line equivalence	Lean-proved	Formal inverse maps
Smooth large-field family	Lean-proved	Actual fields and subgroup domains
Capacity-grid list equality	Lean-proved	Universal $L_m = \binom{n}{k}$
Endpoint fractional criterion	Lean-proved	Target iff $\binom{n}{k} \le \varepsilon^*\lvert\mathbb{F}\rvert$
Next-shell list lower bound	Lean-proved	Explicit $\lvert\mathbb{F}\rvert^m$ injection
Complete list phase split	Lean-proved	Three regimes and exact real interval
Generic minimum-distance interpretation	Lean-proved	Attained minima, covering and good-challenge farness
Pair-MCA certificate bound	Lean-proved	$\lvert\,\mathrm{Bad}_{\mathrm{MCA}}\,\rvert \le \binom{n}{k+1}$
MCA target-fraction inequality	Lean-proved	$\binom{n}{k+1} \le \varepsilon^*\lvert\mathbb{F}\rvert$ for the explicit family
Exact supplied-code MCA grid API	Lean-proved	Operational target iff $2^{128}B_{\mathcal{C}}(r) \le \lvert\mathbb{F}\rvert$, with concrete next-line pinning
Exact RS small-radius certificates	Lean-proved	$B_{\mathcal{C}}(0) = 1$, and $B_{\mathcal{C}}(1) \ge 2$ for $1 \le k,\ k+2 \le n$
Smooth boundary-field instance	Lean-proved	Prime $q = 2^{128} + 192513 \equiv 1 \pmod{4096}$
Boundary-field MCA threshold	Lean-proved	Greatest safe count 0; first unsafe count 1

Obligation	Status	Required endpoint
Explicit-family MCA greatest radius	Lean-proved	Full count radius n for the constructed family
Boundary-field real supremum	Lean-proved	$1/4096$, not attained
Native scalar/pair metric comparison	Lean-proved	$d_C \le d_P \le 2d_C$
Intrinsic modular application	Lean-proved	Actual integral coboundary validity, full original payload and $d_C \le d_P = D_2 \le 2d_C$
Sharp near scaling	Lean-proved	Factor two attained by $(u, 0)$
Canonical raw payload	Lean-proved	All three distances $3/4$ at $(q, n, k) = (12289, 4096, 1024)$
Monomial capacity far family	Lean-proved	Scalar, pair and flattened distances $(n-k)/n$
Explicit challenge profiles	Lean-proved	All-good zero direction; diagonal bad set $\{-1\}$
Arbitrary fixed-mix obstruction	Lean-proved	Capacity-far pair with challenged word zero
Exact-component residual obstruction	Lean-proved, auxiliary	Nonfactorization and no positive far transfer under the stated unrestricted/nonzero-coexact hypotheses
Full projective reconstruction	Lean-proved, auxiliary	Subspace/projector recovery from genuine full-frame data

The counting, calibration, boundary-field and canonical application proofs use only the standard Lean logical axioms `propext`, `Classical.choice`, and `Quot.sound`. The arithmetic certificates are kernel-checked: the boundary primality proof uses an explicit Lucas factor chain, and modular powers are checked by binary modular-exponentiation proof terms. The field, roots and domain parameters are unchanged. Earlier versions used `native_decide` for some of these certificates; their dated dependency reports are historical evidence rather than the trust footprint of the present proofs. This arithmetic change does not construct a commitment protocol or a stronger source-metric comparison.

8.4 Submission boundary

A completed claim requires:

(i) no placeholder assumptions for the main list/MCA result;

(ii) an explicit statement of imported literature theorems;

(iii) all field-size and rounding conditions checked;

(iv) a clear separation between proved theorems, computational evidence, and conjectures;

(v) a reproducible proof artifact and manuscript cross-reference.

8.5 Remaining interfaces

The nine named standalone coding statements now have complete Lean endpoints, including the all-parameter smooth-family realization, complete list phase split, real safe intervals and their excluded suprema, and the generic minimum-distance interpretation. The generic source-to-pair calibration implication

$$d_{\text{source}}(x, \text{valid}) \geq \delta \Longrightarrow \Delta(\pi(x), \mathcal{C}^{\equiv 2}) \geq \delta$$

is Lean-proved for an abstract distance model from its named pointwise non-collapse field. The distance-calibration module identifies the target model with the generic attained pair-code minimum. Pullback distance makes that field reflexive; an application choosing a different pre-existing source metric supplies one explicit comparison. Separately, the intermediate list-decoding regime still requires new ideas if one wants an optimal sub-endpoint radius for a supplied field that fails the binomial target inequality.

The existing commitment-to-plane extraction, canonical plane/pair Hamming isometry, exact transported-code distance equality, and public-observation pullback calibration live in the concrete application hierarchy. They are described in Section 7 and are outside the standalone `ProximityCoding` import closure. Section 7 supplies an explicit modular instance of those interfaces. Comparison with an independently chosen HCP or frame score remains a separate mathematical question.

The explicit source realization lives in the `Sheaf/SystematicCode` modules, including `Modular`. Their actual integral cellular coboundary is proved compatible with RS reduction. The storage modules `CoordinateCode c`, `NoWrapCalibration` and `ThreeLevelMetrics` retain the original payload

and separate the restricted Euclidean comparison from the global native-metric comparison. The canonical specialization and public-observation joins are in `CanonicalSystematic` and `CanonicalThreeLevel`; `CanonicalCalibration` composes the existing interfaces with their declared near/far scales. These modules live under `Protocol/Storage`. Their file location does not change manuscript ownership: Paper 2 specifies the shared construction and source/observation contracts [vdV26b]; this paper proves the coding and quantitative metric comparisons for that exact construction. The two Paper 3 verification configurations separate core and bridge statements for source-size purposes, without introducing another paper or protocol result.

These explicit profiles do not identify arbitrary old HCP presentations or their frame scores with native modular edit distance. A stronger upstream geometric comparison, or a different challenge/observation profile, needs its own theorem. Such obligations remain separate from the completed generic coding results and do not require Paper 2 consensus or Rust.

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